WORKED SOLUTION

STEP III 1996, Question 14

Probability and Statistics • Two-State Markov Process

STEP III 1996, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q14

The key idea

Chop time into tiny intervals of length δt and ask what can happen in one of them. Expected counts then add up by linearity of expectation, and the probability of being mid-repair obeys a simple "in minus out" balance that turns into a first-order linear differential equation as $\delta t \to 0$. Once we have $p(t)$, the expected repair time is just its integral.

Solution

Expected number of punctures

Only time spent actually cycling can produce a puncture, and while I am cycling all is well. Split the total cycling time T into n small pieces of length $\delta t = T/n$. Let X_k be the number of punctures in the kth piece; to first order in δt, X_k is 1 with probability $\alpha\,\delta t$ and 0 otherwise, so $\mathrm{E}(X_k) = \alpha\,\delta t$. By linearity of expectation (no independence needed),

$$\mathrm{E}(\text{number of punctures}) = \sum_{k=1}^{n} \mathrm{E}(X_k) = n\alpha\,\delta t = \alpha T.$$

Time spent repairing doesn't enter, because the clock here is cycling time only.

The equation for $p(t)$

Now let t be real (clock) time, including repairs. At time t I am either repairing (probability $p(t)$) or cycling (probability $1 - p(t)$). I am repairing at time $t + \delta t$ if either

- I was repairing at t and the repair was *not* completed in $(t, t+\delta t)$: probability $p(t)(1 - \beta\,\delta t)$; or
- I was cycling at t and got a puncture in $(t, t+\delta t)$: probability $(1 - p(t))\,\alpha\,\delta t$.

These are exclusive, so, ignoring terms of order $(\delta t)^2$ (two events in one tiny interval),

$$p(t+\delta t) = p(t)(1-\beta\,\delta t) + \big(1-p(t)\big)\alpha\,\delta t.$$

Rearranging,

$$\frac{p(t+\delta t)-p(t)}{\delta t} = \alpha - (\alpha+\beta)\,p(t),$$

and letting $\delta t \to 0$ gives

$$p'(t) = \alpha - (\alpha+\beta)\,p(t).$$

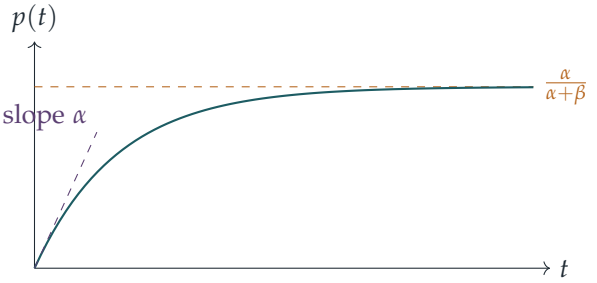
I start with the bike in good order, so the initial condition is $p(0) = 0$.

Checking the given solution

For $p(t) = \dfrac{\alpha}{\alpha+\beta}\big(1-\mathrm{e}^{-(\alpha+\beta)t}\big)$ we have $p(0) = \frac{\alpha}{\alpha+\beta}(1-1) = 0$, and

$$\begin{aligned} p'(t) &= \alpha\,\mathrm{e}^{-(\alpha+\beta)t}, \\ \alpha - (\alpha+\beta)p(t) &= \alpha - \alpha\big(1-\mathrm{e}^{-(\alpha+\beta)t}\big) = \alpha\,\mathrm{e}^{-(\alpha+\beta)t}. \end{aligned}$$

These agree, so p satisfies the differential equation and the initial condition, as required.



The graph shows p rising from 0 with initial slope α towards the long-run level $\frac{\alpha}{\alpha+\beta}$.

Expected repair time

Let R be the time spent mending during $[0,T]$, and let $I(t)$ be 1 if I am repairing at time t and 0 otherwise, so $R = \int_0^T I(t)\,\mathrm{d}t$. Since $\mathrm{E}(I(t)) = p(t)$, taking expectations (again just linearity, applied to the integral as a limit of sums over small intervals) gives

$$\begin{aligned} \mathrm{E}(R) = \int_0^T p(t)\,\mathrm{d}t &= \frac{\alpha}{\alpha+\beta}\left[t + \frac{\mathrm{e}^{-(\alpha+\beta)t}}{\alpha+\beta}\right]_0^T \\ &= \frac{\alpha}{\alpha+\beta}\left(T - \frac{1-\mathrm{e}^{-(\alpha+\beta)T}}{\alpha+\beta}\right). \end{aligned}$$

Ask yourself Why is the expected time mending equal to the integral of a probability?

In each small interval $(t, t+\delta t)$, the time spent mending is about δt with probability $p(t)$ and 0 otherwise, so its expectation is $p(t)\,\delta t$. Adding over all the intervals gives $\sum p(t)\,\delta t \to \int_0^T p(t)\,\mathrm{d}t$.

The two approximations

Write $x = (\alpha+\beta)T$. The expected fraction of the journey spent mending is

$$F = \frac{\mathrm{E}(R)}{T} = \frac{\alpha}{\alpha+\beta}\left(1 - \frac{1-\mathrm{e}^{-x}}{x}\right).$$

x small. Using $\mathrm{e}^{-x} = 1 - x + \frac{1}{2}x^2 - \cdots$, we get $\dfrac{1-\mathrm{e}^{-x}}{x} = 1 - \frac{1}{2}x + O(x^2)$, so

$$F \approx \frac{\alpha}{\alpha+\beta}\cdot\frac{x}{2} = \frac{\alpha}{\alpha+\beta}\cdot\frac{(\alpha+\beta)T}{2} = \frac{\alpha T}{2}.$$

This makes sense: over a short journey $p(t) \approx \alpha t$, which averages to $\alpha T/2$ over $[0, T]$.

x large. Then $0 < \dfrac{1-\mathrm{e}^{-x}}{x} < \dfrac{1}{x}$, which is small, so

$$F \approx \frac{\alpha}{\alpha+\beta}.$$

Over a long journey $p(t)$ has settled at its equilibrium value $\frac{\alpha}{\alpha+\beta}$ for almost all of the time.

Checks

- Equilibrium: setting $p' = 0$ gives $\alpha(1-p) = \beta p$, "rate in equals rate out", so $p = \frac{\alpha}{\alpha+\beta}$, matching the large-$T$ limit.
- If $\beta \to \infty$ (instant repairs) everything mending-related tends to 0, as it should.
- With $\alpha = 0.7$, $\beta = 2.3$, $T = 2.5$ the formula gives $\mathrm{E}(R) \approx 0.5056$. A simulation of 2×10^5 journeys (alternating exponential cycling and repair spells) gave 0.5058, and a simulated puncture count over cycling time 2.5 averaged $1.750 = \alpha T$.

Common slips

- Forgetting the factor $(1 - p(t))$ on the puncture term: you can only puncture while you are cycling.
- Using $p(0) = \frac{\alpha}{\alpha+\beta}$ or leaving the initial condition unstated; the bike starts in good order, so $p(0) = 0$.

- Dropping the second-order term in the small-x expansion, which leaves $F \approx 0$ instead of $\alpha T/2$.

Beyond STEP

This is a two-state continuous-time Markov chain: the bike flips between "working" and "being mended" at rates α and β, and the equation for p is the simplest case of the Kolmogorov forward equations. The same model describes machines that break down and get repaired, ion channels opening and closing, and much of queueing theory. The puncture count in cycling time is a Poisson process with rate α. Grimmett and Welsh, *Probability: An Introduction*, covers Poisson processes and Markov chains at the right level.

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